A Deterministic and Auditable 72-Hour Multi-Airport Digital-Twin Simulation Framework for Operational Disruption Experiments

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Abstract. Airport digital twins are increasingly used for visualization and operational decision support, yet many implementations provide limited evidence on repeatability, rule compliance, and disturbance-isolation capability. This paper presents a deterministic and auditable 72-hour multi-airport simulation framework for a unified airline fleet operating across Hangzhou Xiaoshan, Shanghai Pudong, and Nanjing Lukou airports. The framework combines a frozen scenario snapshot, seed-controlled task generation, resource-constrained discrete-event execution, explicit runway, stand, aircraft, and ground-support constraints, transition guards, invariant audits, and event-log replay fingerprints. A paired five-condition experiment is conducted under normal operation, severe low visibility, temporary runway closure, ground-support-equipment outage, and combined disruption. Twenty seeds (42-61) yield 100 formal runs for a 65-aircraft fleet and approximately 320 missions per run. All runs satisfy the current operational-validity gates and the first seed is deterministically replayed. Relative to normal operation, weather restrictions increase mean resource waiting time by 6.49 h, equipment outages by 41.93 h, and combined disruptions by 59.50 h. Runway closure triggers an average of 66.15 runway reassignments without increasing aggregate resource waiting in the tested capacity regime. The contribution is a reproducible experimental substrate for subsequent multi-agent scheduling and resilience studies, rather than a claim of real-airport predictive validity.

Keywords: Airport digital twin; discrete-event simulation; operational disruption; deterministic replay; verification and validation; ground support equipment.

1 Introduction

Airports are tightly coupled sociotechnical systems in which aircraft rotations, runway movements, stands, maintenance activities, and mobile ground-support resources interact under strict temporal and safety constraints. A local shortage or disruption may propagate through successive missions and across airports. Digital-twin research has therefore moved beyond asset visualization toward operational monitoring, simulation, and decision support. Conde et al. proposed an airport digital-twin conceptualization and data model for turnaround-event management [1]. Postorino et al. used discrete-event simulation to evaluate how airside disruptions propagate across landing, take-off, and turnaround processes [2]. These studies demonstrate the value of integrated operational models, but reproducibility and explicit software-level rule auditing are not always treated as first-class experimental outputs.

A second gap concerns credibility. A systematic review by Souza et al. found that many systems labelled as digital twins implement only a subset of digital-twin capabilities and may remain advanced simulation models [6]. Bitencourt et al. reported

that relatively few digital-twin studies perform both verification and validation using a structured procedure [7]. For simulation-based reliability research, a result should not only be numerically plausible; it should also be traceable to a versioned input, repeatable under controlled randomness, and rejected when operational invariants are violated.

This paper develops a 72-hour, three-airport operational simulation framework with deterministic replay and automated rule auditing. Four questions are studied: whether the scenario produces complete, rule-compliant and replayable traces; how isolated weather, runway, and equipment disruptions affect delay and reassignment; whether combined disruption is stronger than individual disruptions; and which modeled resource is the dominant bottleneck.

The contributions are: (1) a unified three-airport scenario contract for one airline fleet; (2) deterministic event execution and replay auditing; (3) invariant and transition-guard checks coupled to every run; and (4) a paired multi-seed experiment with confidence intervals and standardized within-seed effects.

2 Related Work

2.1 Airport Digital Twins and Operational Simulation

Airport digital-twin work covers information integration, turnaround management, air-traffic visualization, and resource optimization. Conde et al. connected heterogeneous airport information through a FIWARE-based architecture [1]. Chen et al. investigated a tangible mixed-reality digital twin for collaborative air-traffic-management operations [5]. Those studies strengthen visualization and situational awareness; the present study instead emphasizes repeatable experiment execution and machine-checkable operational traces.

Disruption modeling has also received sustained attention. Postorino et al. integrated landing and take-off processes with turnaround activities to capture knock-on effects [2]. Ground handling is sensitive to scarce mobile resources. Saggar et al. proposed a real-time booking mechanism for ground-support equipment under uncertainty [3], while Sahadevan et al. combined arrival-time prediction with allocation of limited ground resources [4]. These findings motivate explicit ground-support capacity rather than an exogenous turnaround-delay scalar.

2.2 Digital-Twin Credibility and Reproducibility

Digital twins integrate models, data, and decision processes, but terminology and validation practice remain inconsistent. Fuller et al. surveyed enabling technologies and open challenges [8]. Souza et al. distinguished simulation capability from fuller digital-twin capability [6]. Bitencourt et al. highlighted the scarcity of structured verification and validation practice [7]. This work separates software verification from empirical validation. Replay, invariants, and transition guards support internal credibility; real operational calibration remains future work.

3 Framework Design

3.1 Scenario Contract

The simulation accepts a frozen scenario snapshot containing three airports, a single 65-aircraft fleet, aircraft health attributes, stands, runway profiles, ground-support vehicles, and route resources. Each run covers 72 hours. The default planner limits each aircraft to five missions, uses a simulated fault probability of 0.025 and emergency probability of 0.01, enforces 180-s runway separation, assumes 30-min ground-support service, and permits six parallel support teams.

Every mission references one aircraft, origin and destination airports, departure and arrival resources, and an operational event sequence. The same snapshot and seed generate the same base mission population before experimental disturbances are applied.

3.2 Resource-Constrained Execution

A discrete-event engine orders events by simulation time and deterministic tie-breaking keys. Runway, stand, aircraft, maintenance, and ground-support availability are explicit constraints. When a requested resource is unavailable, the engine may reassign a compatible resource or delay the dependent event. A state transition is committed only after its guard rules pass. Intended and realized assignments are both retained, so exposure and response can be distinguished.

3.3 Audit and Replay Layers

Each run passes three credibility gates: end-to-end acceptance checks complete mission chains; invariant auditing checks resource exclusivity, state consistency and accounting identities; transition guarding rejects lifecycle violations. For deterministic replay, the same input and seed are executed twice. Ordered event logs are reduced to versioned checkpoints and a fingerprint. Replay passes only when event sequence, checkpoint states and final metrics agree.

4 Experimental Method

4.1 Paired Design

Table 1. Experimental conditions.

Condition	Disturbance definition
Baseline	No injected weather, runway, or external equipment disruption.
Weather only	Severe low visibility at all airports from hour 0 to 18; runway-capacity factor 0.60 and ground-speed factor 0.65.
Runway closure	Shorter formal runway at each airport closed from hour 4 to 14 for inspection.
Equipment outage	First available unit in each support category at each airport unavailable from hour 0 to 14.
Combined	Weather, runway closure, and equipment outage occur together.

Twenty seeds (42-61) are used. Within each seed, all five conditions share the same snapshot, planning parameters, and base task-generation randomness. Differences from baseline are paired within seed. The experiment contains 100 formal runs. Seed 42 is

additionally repeated for deterministic replay and replay runs are not counted as independent samples.

4.2 Measures and Analysis

Primary outcomes are mission completion rate, operational delay rate, total resource waiting hours, average and 95th-percentile resource wait, and fleet utilization. Mechanism indicators include weather-affected missions, runway reassignments, and equipment reassignments. A run is valid only when acceptance is accepted, invariant status is passed, and transition-guard status is passed.

For outcome y, the within-seed effect is $d_{c,s} = y_{c,s} - y_{0,s}$. We report the mean difference, a Student-t 95% confidence interval, and the paired standardized effect d_z = mean(d)/SD(d).

5 Results

All 100 formal runs passed the implemented operational-validity gates, and the five conditions for seed 42 were deterministically replayed. Each run planned approximately 320 missions and mean completion was 1.00 under all conditions.

Table 2. Results across 20 paired seeds.

Condition	Delay	Wait h	95% CI	Delta h	dz
Baseline	0.3398	113.70	93.42-133.99	0.00	0.00
Weather only	0.3566	120.19	99.67-140.71	6.49	1.73
Runway closure	0.3398	113.70	93.42-133.99	0.00	0.00
Equipment outage	0.3894	155.63	133.02-178.24	41.93	3.37
Combined	0.4142	173.20	149.99-196.41	59.50	4.31

Weather restrictions affect a mean of 91.75 missions and increase resource waiting by 6.49 h relative to paired baselines (95% CI 4.73-8.25). Equipment outages produce 271.85 equipment reassignments on average and increase waiting by 41.93 h (95% CI 36.10-47.76). Combined disruption has the largest observed effect: a 59.50-h increase (95% CI 53.03-65.96) and an operational delay rate of 0.4142.

Runway closure yields 66.15 runway reassignments per run but no aggregate waiting-time increase. This does not imply operational irrelevance. The closure mechanism is activated while modeled spare capacity absorbs the selected closure, illustrating why mechanism and outcome indicators must be reported together.

6 Discussion

6.1 Answers to the Research Questions

For RQ1, the implementation produces complete 72-hour traces that pass the current rule gates and can be deterministically replayed. This establishes internal consistency, not real-world predictive validity. For RQ2, isolated disturbances produce different signatures: weather reduces capacity and speed; runway closure causes substitution; and equipment outage creates repeated support-resource reassignment and substantial waiting. For RQ3, combined disruption has the highest delay rate and waiting time, although a full factorial design is required before claiming nonlinear interaction. For RQ4, ground-support availability is the dominant modeled bottleneck under the selected parameters.

6.2 Reliability Significance

The framework represents external stressors and traces their effects through constrained mission chains. It also treats the simulator as a system requiring credibility evidence. A result is retained together with its scenario identifier, seed, event fingerprint, rule-audit outcome, and paired statistics. This converts a visual demonstration into a reproducible experiment.

The framework is also a precondition for later multi-agent optimization. Future agents can propose runway, stand, maintenance, or support decisions, while existing guards and invariants define a feasibility envelope. The present results are therefore a controlled baseline before agent policies are introduced.

7 Threats to Validity

Construct validity. Waiting and delay represent selected resilience dimensions but exclude passenger impact, crew legality, fuel, emissions, and financial loss.

Internal validity. The paired-seed design isolates disturbances under a fixed implementation. The combined experiment is not a full factorial design, so statistical interaction is not yet identified.

External validity. Disturbances are synthetic and the model is not calibrated against historical records from the three airports. Results are not forecasts of actual airport performance.

Model scope. External traffic, air-traffic-control decisions, passenger connections, crew rotations, and detailed vehicle-routing conflicts remain outside the experiment.

Statistical conclusion validity. Twenty paired seeds provide initial conference-scale evidence, but broader parameter sweeps and more seeds are required for sensitivity analysis.

8 Conclusion and Future Work

This paper presented a deterministic and auditable 72-hour multi-airport digital-twin simulation framework. A 65-aircraft fleet was simulated across three airports under five paired conditions and 20 seeds. Weather increases delay moderately, runway closure is absorbed through reassignment in the tested capacity regime, equipment outage creates the largest single-resource bottleneck, and combined disruptions have the strongest overall effect.

Future work will add per-airport propagation metrics, recovery-time measures, and comparison policies such as first-come-first-served, static assignment, and rule-based dynamic reassignment. Historical or semi-physical data will be used for calibration. Multi-agent policies can then be evaluated against the same frozen snapshots, paired seeds, and audit gates.

References

[1] Conde, J., et al.: Applying digital twins for the management of information in turnaround event operations in commercial airports. Advanced Engineering Informatics 54, 101723 (2022). doi:10.1016/j.aei.2022.101723.

[2] Postorino, M.N., Mantecchini, L., Malandri, C., Paganelli, F.: A methodological framework to evaluate the impact of disruptions on airport turnaround operations: A case study. Case Studies on Transport Policy 8(2), 429-439 (2020). doi:10.1016/j.cstp.2020.03.007.

[3] Saggar, S., Tomasella, M., Cattaneo, G., Matta, A.: Enhanced operational management of airport ground support equipment for better aircraft turnaround performance. Winter Simulation Conference (2021).

[4] Sahadevan, D., Al Ali, H., Notman, D., Mukandavire, Z.: Optimising airport ground resource allocation for multiple aircraft using machine learning-based arrival time prediction. Aerospace 10(6), 509 (2023). doi:10.3390/aerospace10060509.

[5] Chen, K., Nadirsha, T.N.M., Lilith, N., Alam, S., Svensson, A.: Tangible digital twin with shared visualization for collaborative air traffic management operations. Transportation Research Part C 162, 104546 (2024). doi:10.1016/j.trc.2024.104546.

[6] Souza, V., Cruz, R., Silva, M., Lins, S., Lucena, V.: When is a simulation a digital twin? A systematic literature review. Manufacturing Letters 35, 940-951 (2023). doi:10.1016/j.mfglet.2023.08.014.

[7] Bitencourt, J., Wooley, A., Harris, G.: Verification and validation of digital twins: A systematic literature review for manufacturing applications. International Journal of Production Research 63(1), 342-370 (2025). doi:10.1080/00207543.2024.2357741.

[8] Fuller, A., Fan, Z., Day, C., Barlow, C.: Digital twin: Enabling technologies, challenges and open research. IEEE Access 8, 108952-108971 (2020). doi:10.1109/ACCESS.2020.2998358.